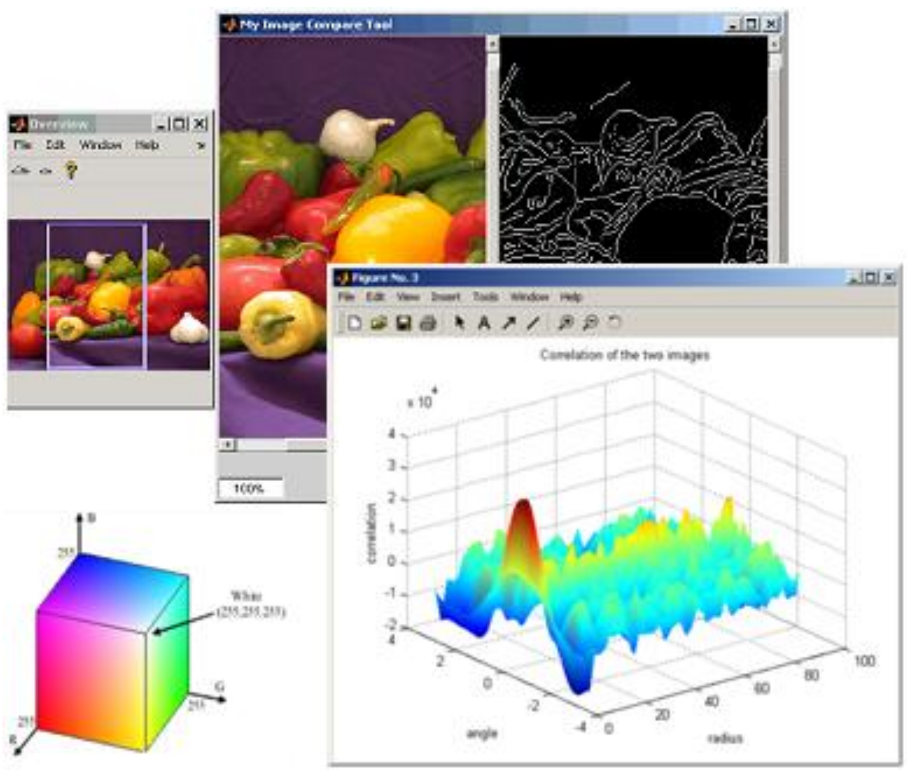
Digital Image Processing

LECTURE 7: IMAGE ARITHMETIC AND LOGICAL OPERATION



Dr. Ajay Kumar Mahato
Assistant Professor
ECE Department
GLA University Mathura

Arithmetic Operation on Images

Image Arithmetic Operation

- Arithmetic operations include image addition, subtraction, multiplication, division and blending.

Image Addition	$g(x,y) = f_1(x,y) + f_2(x,y)$
Image Subtraction	$g(x,y) = f_1(x,y) - f_2(x,y)$
Image Multiplication	$g(x,y) = f_1(x,y) \times f_2(x,y)$
Image Division	$g(x,y) = \dfrac{f_1(x,y)}{f_2(x,y)}$
Image Blending	$g(x,y) = \alpha \times f_1(x,y) + (1-\alpha)f_2(x,y)$

Arithmetic Operations

Addition:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} a+p & b+q \\ c+r & d+s \end{bmatrix}$$

Subtraction:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} a-p & b-q \\ c-r & d-s \end{bmatrix}$$

Arithmetic Operations

❑ Multiplication Operation:

Array Multiplication:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} .* \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap & bq \\ cr & ds \end{bmatrix}$$

Matrix Multiplication:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} * \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix}$$

In Image Processing, we performed Array Multiplication

Arithmetic Operations

❑ Multiplication Operation:

Array Division:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} ./ \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} a/p & b/q \\ c/r & d/s \end{bmatrix}$$

Matrix Division:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} / \begin{bmatrix} p & q \\ r & s \end{bmatrix} = [A][P]^{-1}$$

In Image Processing, we performed Array Division

Numerical Questions

Ques 1: Consider the following two images.

$$F_1 = \begin{array}{|c|c|c|} \hline 1 & 3 & 7 \\ \hline 5 & 15 & 75 \\ \hline 200 & 50 & 150 \\ \hline \end{array} \qquad F_2 = \begin{array}{|c|c|c|} \hline 50 & 150 & 125 \\ \hline 45 & 55 & 155 \\ \hline 200 & 50 & 75 \\ \hline \end{array}$$

Perform the Addition, Subtraction, Multiplication, Division and image blending operations. Assume both the images are of the 8-bit integer types. (uint8 of MATLAB type.)

Numerical Questions

$$F_1 = \begin{array}{|c|c|c|}\hline 1 & 3 & 7 \\ \hline 5 & 15 & 75 \\ \hline 200 & 50 & 150 \\ \hline \end{array} \qquad F_2 = \begin{array}{|c|c|c|}\hline 50 & 150 & 125 \\ \hline 45 & 55 & 155 \\ \hline 200 & 50 & 75 \\ \hline \end{array}$$

SUM

$$F_1 + F_2 = \begin{array}{|c|c|c|}\hline 51 & 153 & 132 \\ \hline 50 & 70 & 180 \\ \hline 400 & 100 & 225 \\ \hline \end{array} = \begin{array}{|c|c|c|}\hline 51 & 153 & 132 \\ \hline 50 & 70 & 233 \\ \hline 255 & 100 & 225 \\ \hline \end{array}$$

DIFF

$$F_1 - F_2 = \begin{array}{|c|c|c|}\hline -49 & -147 & -118 \\ \hline -40 & -40 & -80 \\ \hline 0 & 0 & 75 \\ \hline \end{array} = \begin{array}{|c|c|c|}\hline 0 & 0 & 0 \\ \hline 0 & 0 & 0 \\ \hline 0 & 0 & 75 \\ \hline \end{array}$$

Numerical Questions

$F_1 =$

1	3	7
5	15	75
200	50	150

$F_2 =$

50	150	125
45	55	155
200	50	75

Mult

$F_1 \times F_2 =$

50	450	875
225	825	11625
40000	2500	11250

$=$

50	255	255
225	255	255
255	255	255

DIV

$F_1 / F_2 =$

0.02	0.02	0.06
0.11	0.27	0.48
1	1	2

$=$

0	0	0
0	0	0
1	1	2

Numerical Questions

$F_1 =$

1	3	7
5	15	75
200	50	150

$F_2 =$

50	150	125
45	55	155
200	50	75

For Blending Operation, Assume $\alpha = 0.6$

$\alpha \times F_1 =$

1	2	4
3	9	45
120	30	90

$(1-\alpha) \times F_2 =$

20	60	50
18	22	62
80	20	30

$\alpha \times F_1 + (1-\alpha) \times F_2 =$

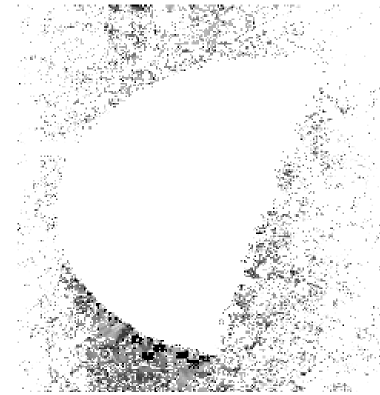
21	62	54
21	31	107
200	50	120

Image Arithmetic Operation

sum=A+B

diff=A-B

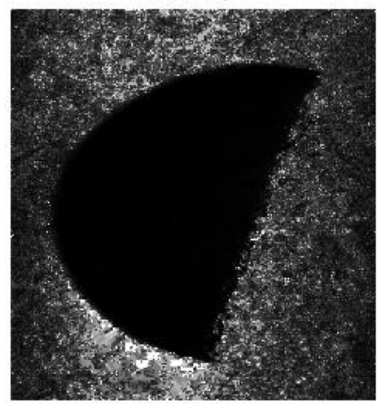
mul=A.*B



Original Image A1

Original Image B1

div=A./B



blend=alpha*A+(1-alpha)*B

alpha = 0.6

Gray Image A

Resize Image B

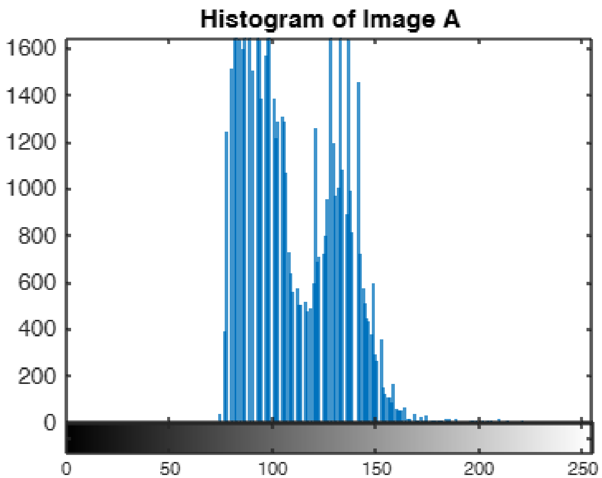
Arithmetic Operations

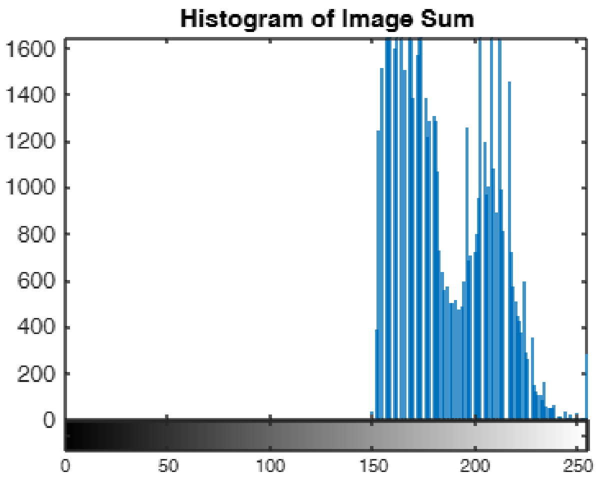
❑ Addition Operations:

Original Image A

+ 75 =

sum=A+75

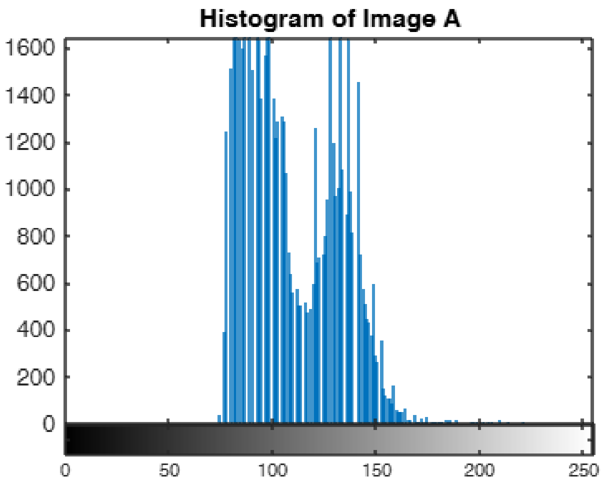


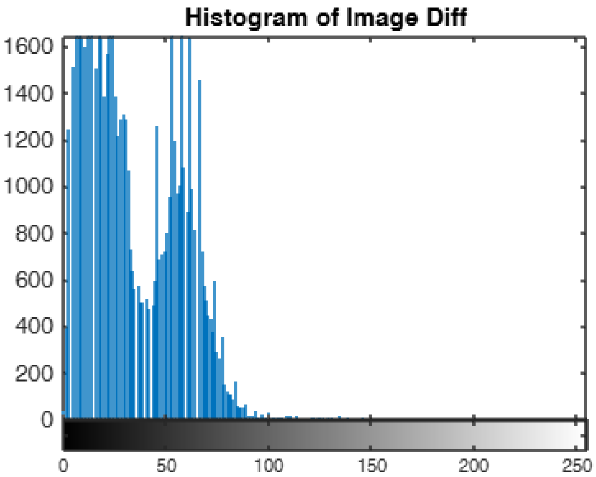


Arithmetic Operations

❑ Subtraction Operation:

- 75 =



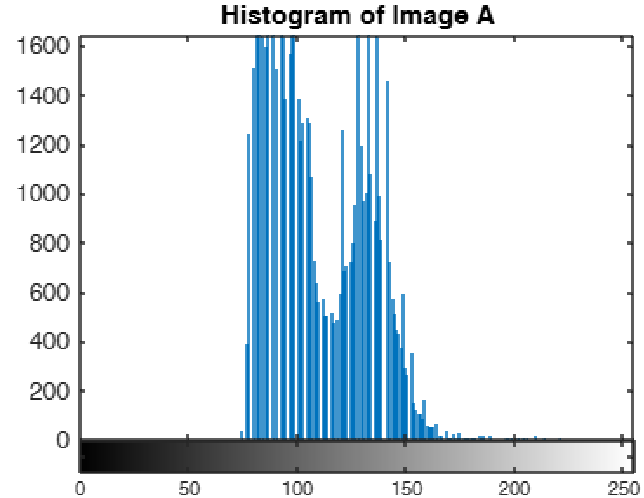


Arithmetic Operations

❑ Multiplication Operation:

Input

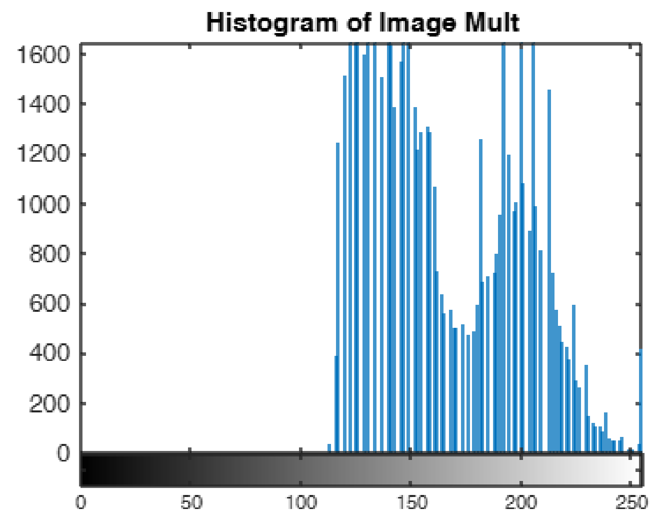
Original Image A



Output

Mult = A*1.5

Mult=A*1.5

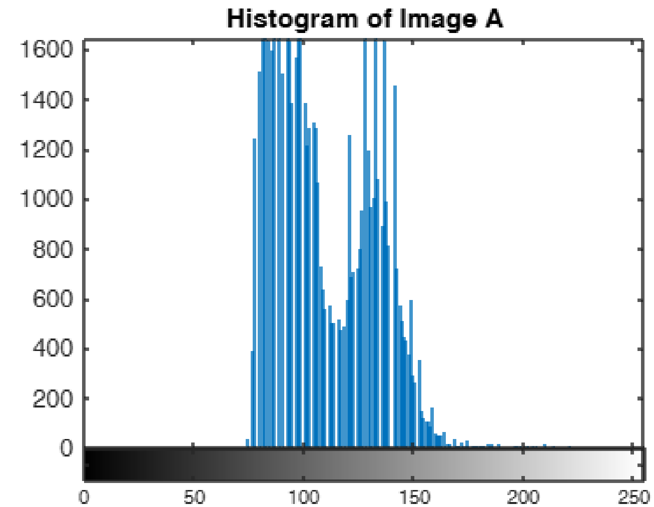


Arithmetic Operations

❑ Multiplication Operation:

Input

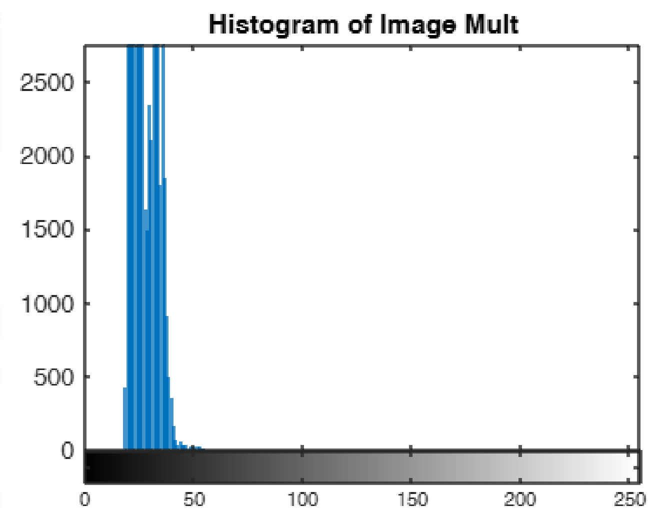
Original Image A



Output

Mult = A*0.25

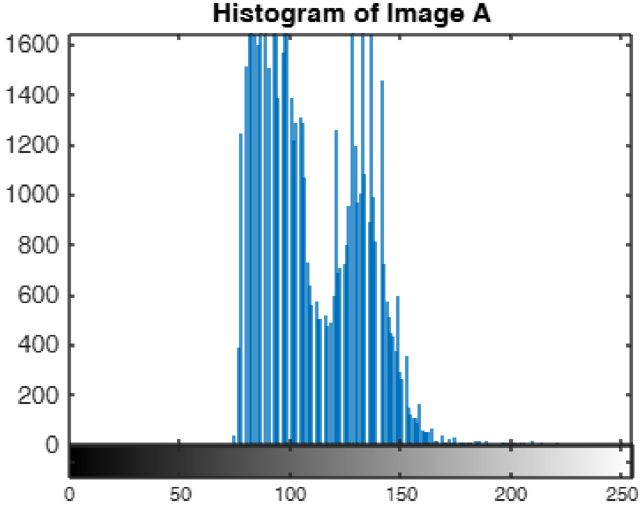
Mult=A*0.25



Arithmetic Operations

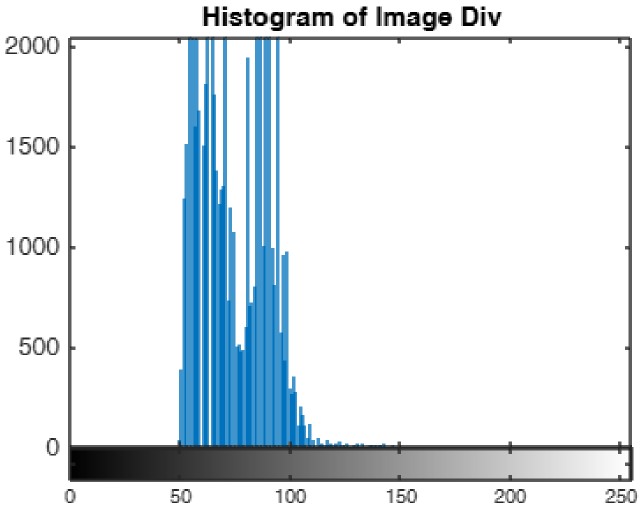
❑ Division Operation:

Input



Output

B=A/1.5



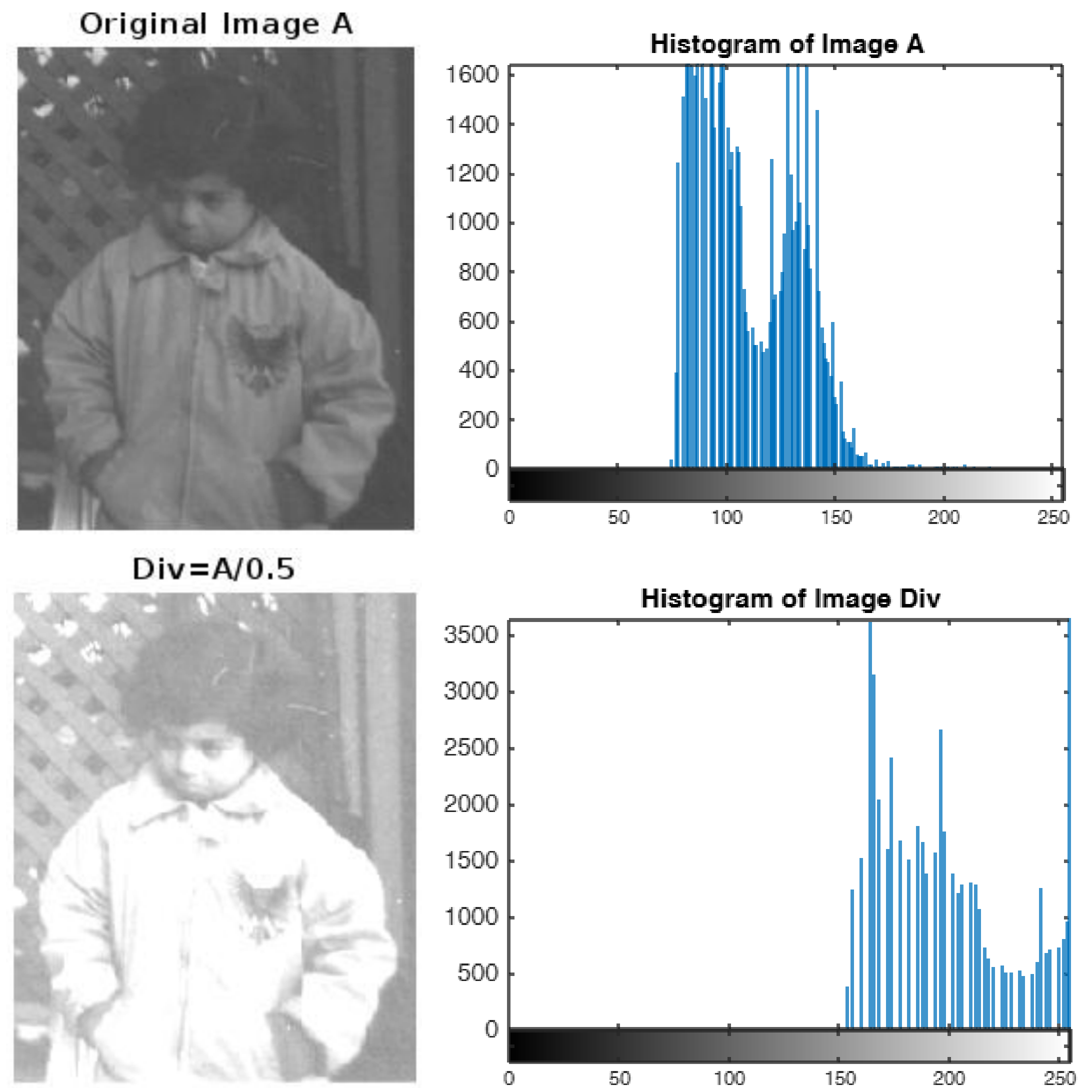
Arithmetic Operations

❑ Division Operation:

Input

Output

B=A/0.5



Arithmetic Operation Applications

- **Image Addition:**
  - Superimposing an image on another image.
  - Increase brightness of an image.
- **Image Subtraction:**
  - Background Elimination
  - Brightness Reduction
  - Change Detection
- **Image Multiplication:**
  - It increases contrast.
  - It is useful for designing filter mask.
  - It is useful for creating a mask to highlight the area of interest.

Arithmetic Operation Applications

- **Image Division:**
  - Contrast reduction
  - Change detection
  - Separation of luminance and reflectance components
- **Image Blending:**
  - It is useful for transparency and compositing.

Logical Operation on Images

Image Logical Operations

- Bitwise operations can be applied to image pixels. Some of the logical operations that are widely used in image processing are as follows:

  - AND Operation
  - EX-OR Operation
  - OR Operation
  - NAND Operation
  - EX-NOR Operation
  - NOR Operation
  - Invert/Logical NOT Operation

Numerical Questions

Ques 2: Consider the following two images.

$$F_1 = \begin{array}{|c|c|c|} \hline 1 & 0 & 0 \\ \hline 1 & 1 & 1 \\ \hline 0 & 0 & 1 \\ \hline \end{array} \qquad F_2 = \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

Perform the logical AND, OR, NOT, and difference operations.

Logical Operations

Image BW1

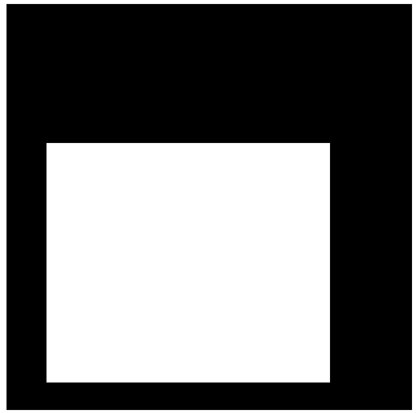
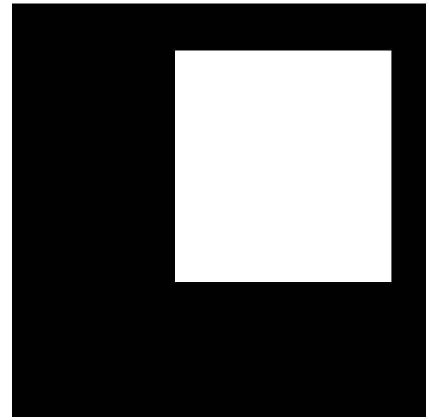
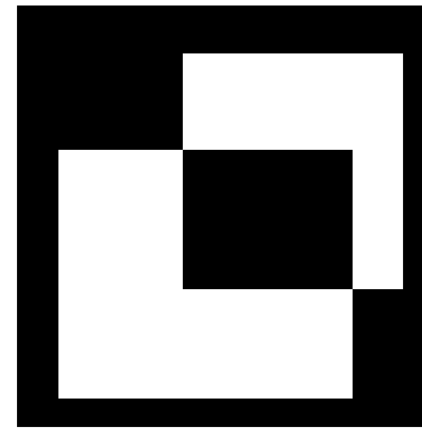


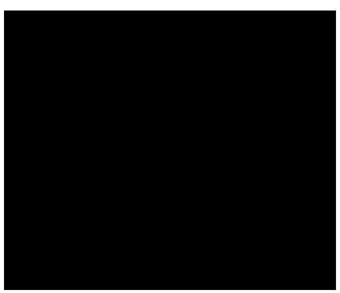
Image BW2



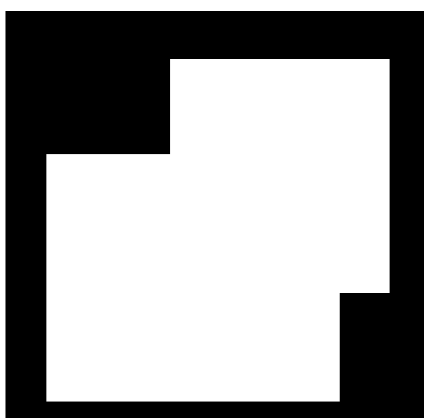
BW1 xor BW2



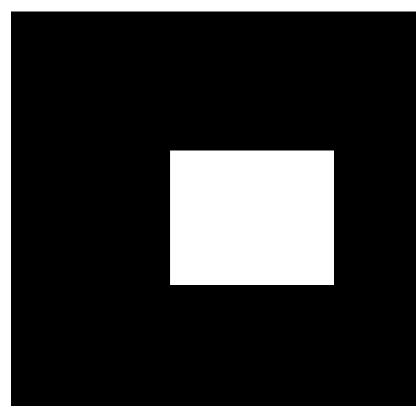
Complement of BW1



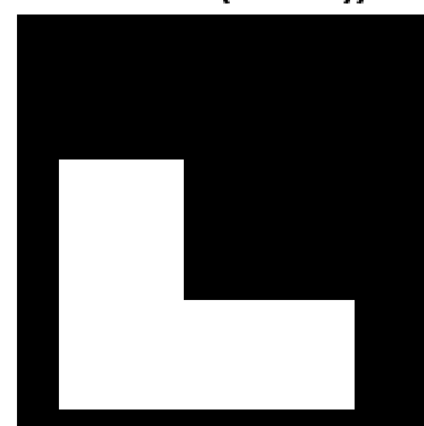
BW1 or BW2



BW1 and BW2



BW1 & (~BW2))



Logical Operations Applications

- **AND and NAND operation:**
  - Computation of the intersection of images
  - Design of filter masks.
  - Slicing of Gray scale images.

- **OR and NOR operation:**
  - OR is used as union operator of two images.
  - OR can be used as a merging operator.

Logical Operations Applications

- **XOR and XNOR Operation:**
  - Change Detection.

- **Invert/Logical NOT Operation:**
  - Obtaining the negative of an image.
  - Making feature clear to the observer.
  - Morphological Processing.

Numerical Questions

Ques 3: Consider the following 4×4, 8 level images A and B. Find A + B, A - B, $A \times B$ and A/B

A =

1	2	3	4
5	5	6	6
6	7	6	6
6	7	2	3

B =

1	3	5	7
8	7	0	1
3	5	6	7
1	3	5	7

Numerical Questions

Ques 4: Consider the following two set of images. The images on the right are the result of applying arithmetic operations between the two images on the left and in the middle. Specify which arithmetic operation has been used on Set1 and Set2.

Hint: Division operation has not been used on any set. Give your answer by filling up the following blanks.

Set1: ....

Set2: ....

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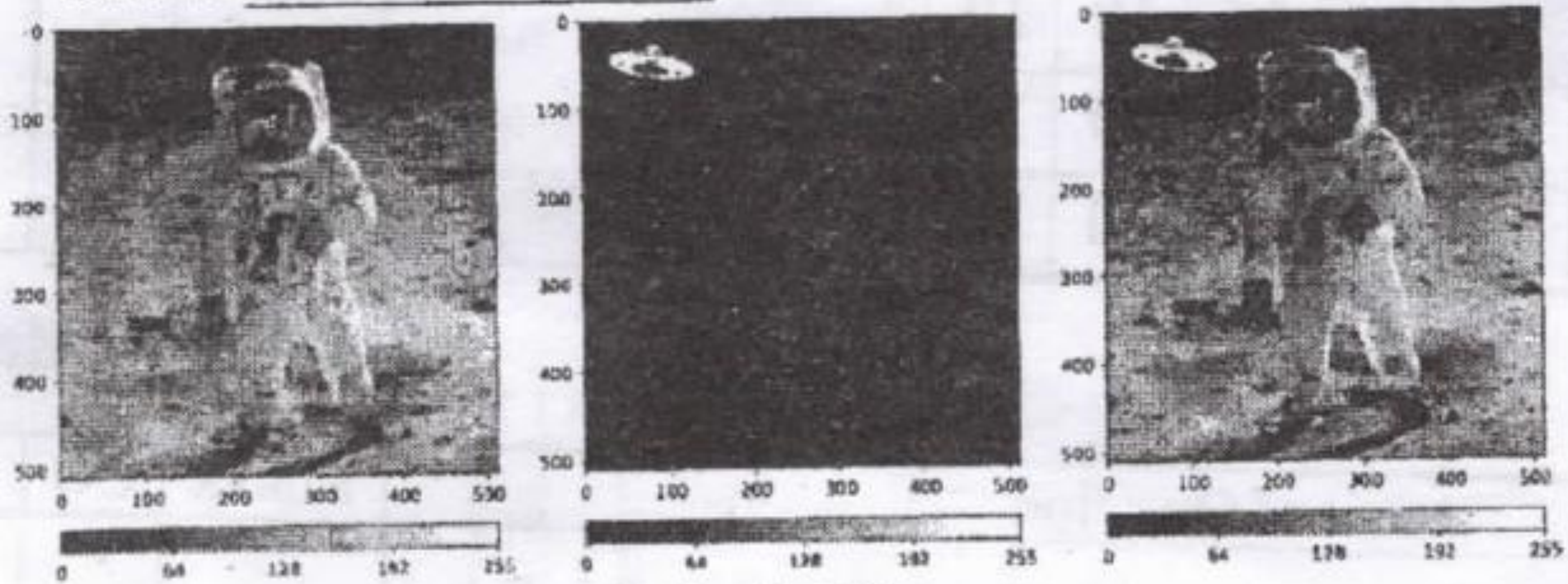
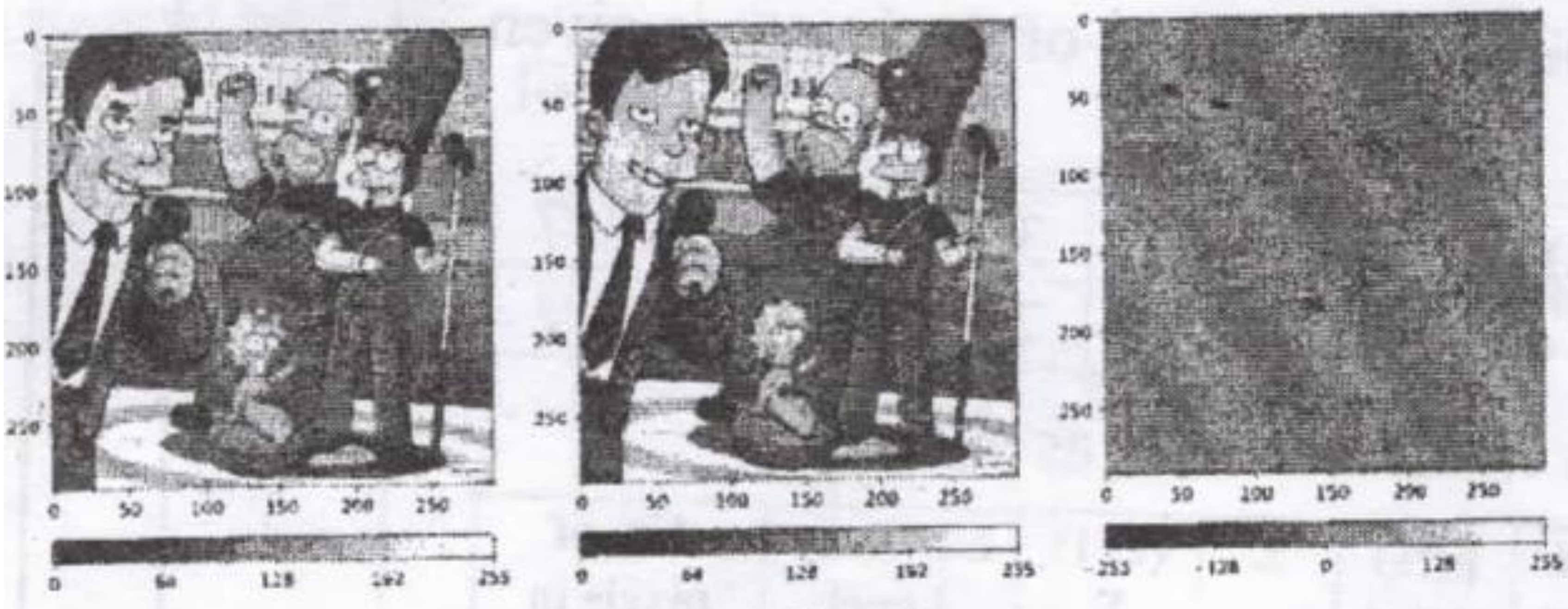


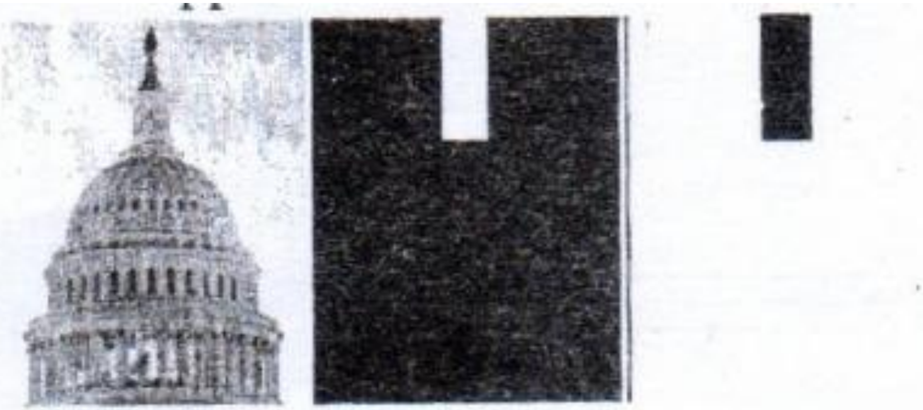
Set 1



Set 2

Numerical Questions

Ques 5: On the original image, an AND Mask and an OR Mask were applied as shown below:



(a) Original image (b) AND image mask (c) OR image mask

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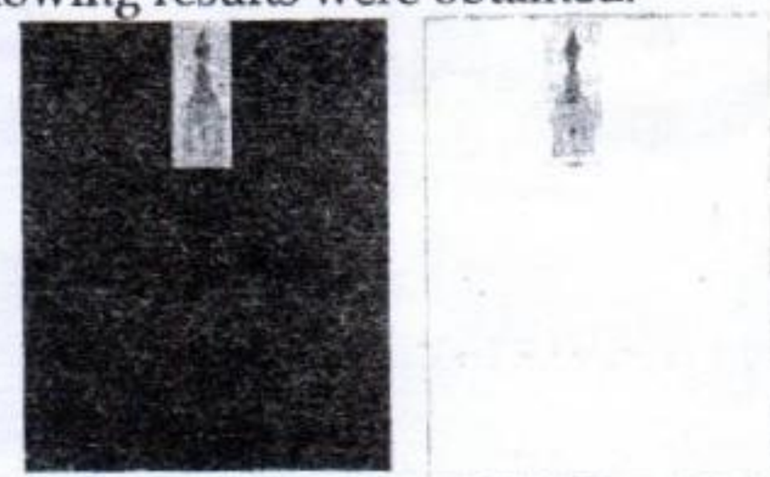
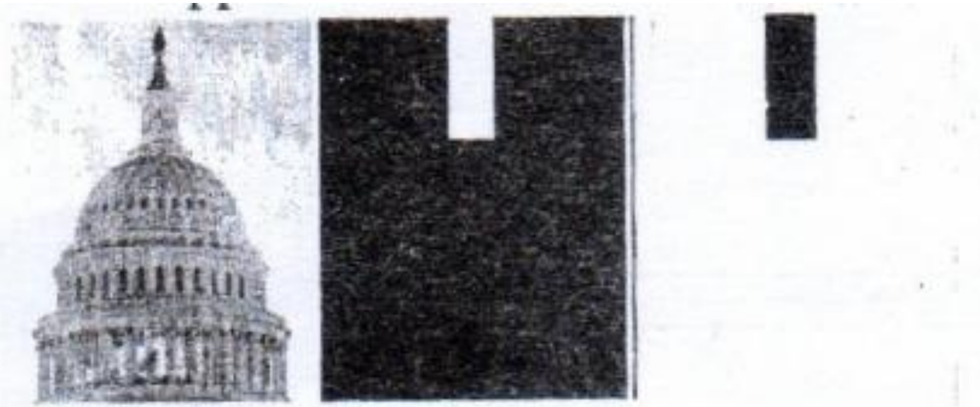
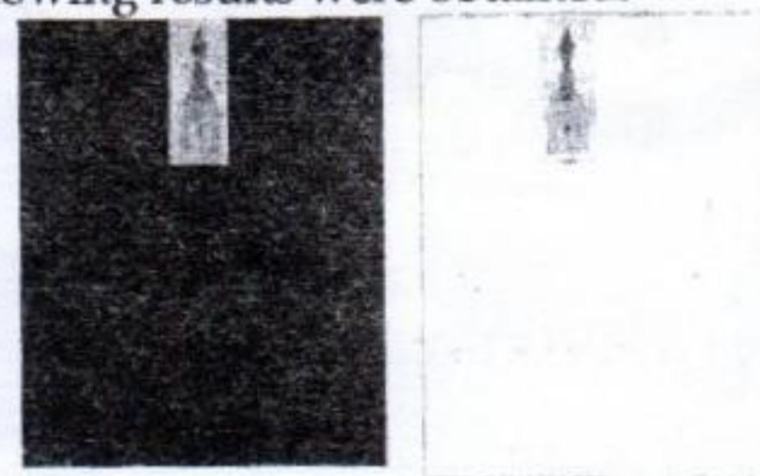
The following results were obtained.





(a) Original image (b) AND image mask (c) OR image mask

The following results were obtained.



Complete the Following Sentences:

The result on the left was obtained by applying .... Mask and the result on the right was obtained by applying .... Mask.

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Numerical Questions

Ques 6: Consider the following two images

$$F_1 = \begin{bmatrix} 100 & 150 & 70 \\ 50 & 150 & 175 \\ 200 & 50 & 150 \end{bmatrix} \quad \text{and } F_2 = \begin{bmatrix} 50 & 50 & 25 \\ 45 & 55 & 50 \\ 50 & 50 & 75 \end{bmatrix}$$

Find $F_1 + F_2$ and $F_1 - F_2$

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Numerical Questions

Ques 7: Find the values of logical operation XOR and Set operations A – B for the binary images A and B. Assume 1 to be the foreground pixels.

0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	0
0	0	0	0	1	1	1	0
0	0	0	0	1	1	1	0
0	0	0	0	0	0	0	0

A

0	0	0	0	0	0	0	0
0	0	1	1	1	0	0	0
0	0	1	1	1	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

B

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Some Commercial Applications

Commercial Applications

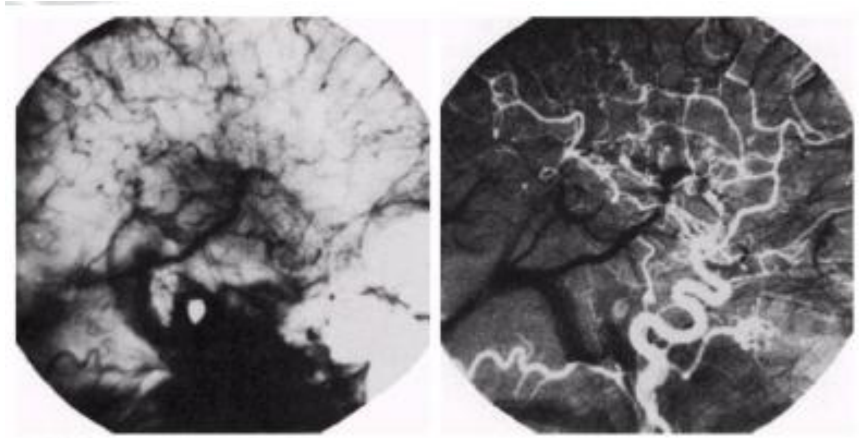
❑ Image subtraction:

One of the most successful commercial applications of image subtraction is **Mask Mode Radiography**

Example: Blood stream is injected with a dye and X-ray images are taken before and after the injection

f(x, y): image after injecting a dye

h(x, y): image before injecting the dye



before injection after injection

Commercial Applications

- **Image Multiplication:**

  One of the most successful commercial applications of image multiplication is **Region of Interest**

Commercial Applications

❑ Image Division:

One of the most successful commercial applications of image division is **Shading Correction**

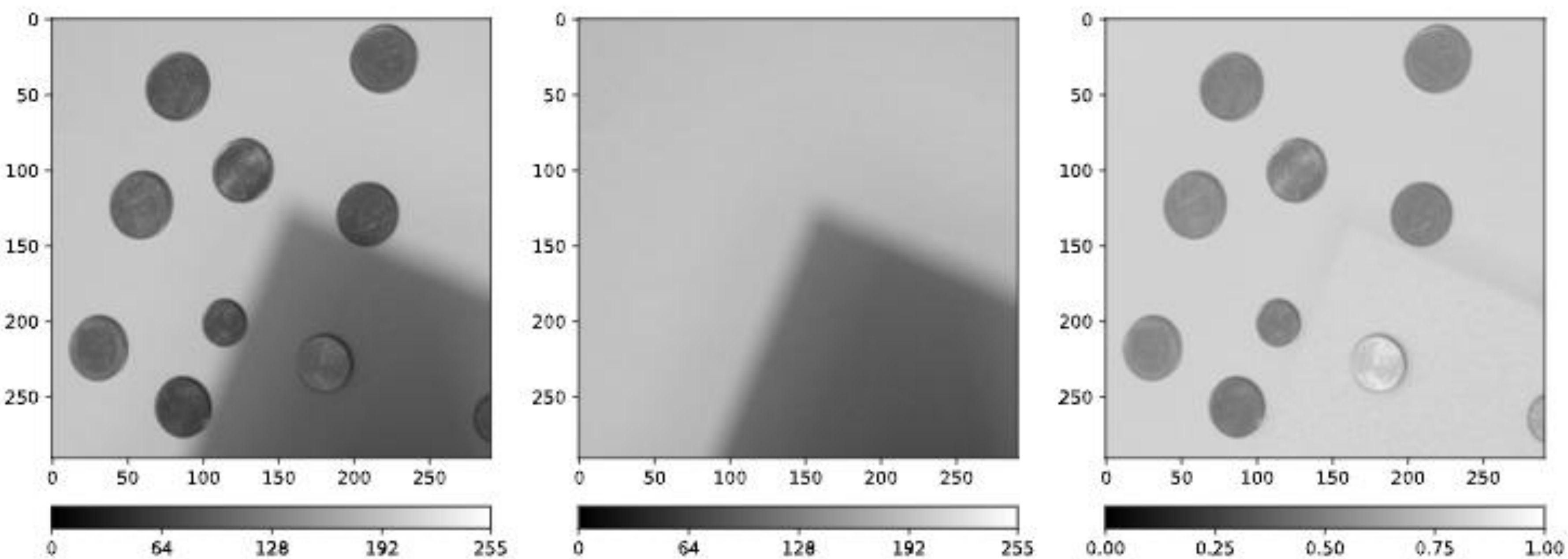


Fig. 19 The right image is the division of the left image by the right image.

Commercial Applications

❑ Image Division:

One of the most successful commercial applications of image division is **Shading Correction**

- An imaging sensor produces image g(x, y),

$$g(x, y) = f(x, y)\, h(x, y)$$

Perfect Image

Shadowing function

- By dividing g(x, y) by h(x, y), we can get the perfect image

$$f(x,y) = \frac{g(x,y)}{h(x,y)}$$

Commercial Applications

❑ Image Arithmetic operation:

One of the most successful commercial applications of image arithmetic operation is **Image Averaging.**

- A noisy image:

$$\boldsymbol{g(x,y) = f(x,y) + n(x,y)}$$

- Averaging M different noisy images:

$$\bar{g}(x,y) = \frac{1}{M}\sum_{i=1}^{M} g_i\,(x,y)$$

- As M increases, the variability of the pixel values at each location decreases
- This means that g(x, y) approaches f(x, y) as the number of noisy images used in the averaging process increases

Thank You